

AYE

TECH HUB

CAD & DESIGN
ENGINEERING GUIDE

AutoCAD Commands Cheat Sheet

2D Drafting · 3D Modeling · Layers · Dimensioning · Blocks
Plotting · Advanced Tools · Shortcuts · Best Practices

200+

COMMANDS

50+

SHORTCUTS

15

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Awet G. Nway

Founder, AYE Tech Hub | CAD & Engineering Design Expert

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SECTION 1
 LINE · CIRCLE · ARC · RECTANGLE · POLYGON

Drawing & Object Commands

These are the foundational drawing commands used in every AutoCAD 2D project. Mastering their options dramatically increases drafting speed.

2D Drawing Commands

Command	Alias	Description & Key Options
LINE	L	Draw straight lines. Options: Close (C), Undo (U). Multi-segment: keep clicking points.
CIRCLE	C	Draw circles. Options: 3P (3-point), 2P (2-point), TTR (tangent-tangent-radius), TTT.
ARC	A	Draw arcs. 11 subtypes: 3-Point, Start-Center-End, Start-Center-Angle, Start-End-Radius.
RECTANGLE	REC	Draw rectangles. Options: Chamfer (C), Elevation (E), Fillet (F), Thickness (T), Width (W).
POLYGON	POL	Draw regular polygons 3–1024 sides. Options: Inscribed in circle / Circumscribed about circle.
ELLIPSE	EL	Draw ellipses. Options: Arc (creates elliptical arc), Center, Axis-End.
PLINE	PL	Draw polylines (connected line+arc segments). Assigns width. Use for closed shapes.
SPLINE	SPL	Draw smooth NURBS curves through or near control points. Types: Fit, CV.
POINT	PO	Place a point object. Display controlled by PDMODE/PDSIZE system variables.
DONUT	DO	Draw filled rings or solid filled circles. Specify inside/outside diameter.
XLINE	XL	Draw infinite construction lines (rays through 2 points). Great for layout.
RAY	RAY	Draw semi-infinite construction ray from a start point through a second point.
SOLID	SO	Draw filled triangular or quadrilateral solid shapes (not 3D solid).
BOUNDARY	BO	Create a closed boundary polyline or region from enclosed areas automatically.
REGION	REG	Convert 2D closed objects into 2D surface regions for Boolean operations.
REVCLOUD	REVCLOUD	Draw revision clouds to mark changed areas on drawings. Options: Arc length, Style.
SKETCH	SKETCH	Freehand sketching with incremental line segments. Rarely used in production.

Object Snap (OSNAP) Settings

OSNAP Mode	Shortcut	Description
Endpoint	END	Snap to the nearest endpoint of lines, arcs, polylines.

Midpoint	MID	Snap to the midpoint of lines, arcs, polylines.
Center	CEN	Snap to the center of circles, arcs, ellipses.
Node	NOD	Snap to point objects.
Quadrant	QUA	Snap to 0°, 90°, 180°, 270° of circles/arcs/ellipses.
Intersection	INT	Snap to the intersection of two objects.
Extension	EXT	Snap to the imaginary extension of a line or arc.
Insertion	INS	Snap to the insertion point of blocks and text.
Perpendicular	PER	Snap perpendicular to a line, arc, or circle.
Tangent	TAN	Snap tangent to a circle or arc.
Nearest	NEA	Snap to the nearest point on any object.
Apparent Intersection	APPINT	Snap where two objects would intersect if extended.
Parallel	PAR	Snap to draw parallel to an existing line.

★ **Pro Tip: OSNAP Keyboard Override**

Type the 3-letter OSNAP shortcut during any command to snap exactly once.

Example: While in LINE command, type MID then click a line → snaps to midpoint.

F3 toggles running OSNAP on/off. ~SHIFT+right-click opens OSNAP context menu.

SECTION 2
 MOVE · COPY · TRIM · EXTEND · OFFSET

Modify & Edit Commands

Modify commands transform, restructure, and refine existing geometry. These are used constantly in professional drafting — mastering them is essential.

Command	Alias	Description & Key Options
MOVE	M	Move objects from one location to another. Uses base point + displacement.
COPY	CO / CP	Copy objects. Options: Array (A) for multiple copies at set spacing.
ROTATE	RO	Rotate objects around a base point. Options: Copy (C), Reference (R).
SCALE	SC	Scale objects uniformly or non-uniformly. Options: Copy (C), Reference (R).
MIRROR	MI	Mirror objects across a line. Option to delete or keep original.
OFFSET	O	Create parallel copies at a specified distance. Options: Through (T), Erase (E).
TRIM	TR	Trim objects to cutting edges. SHIFT+click extends instead of trims.
EXTEND	EX	Extend objects to boundary edges. SHIFT+click trims instead of extends.
FILLET	F	Create rounded corners. R sets radius. Options: Polyline (P), Trim (T), Multiple (M).
CHAMFER	CHA	Create beveled corners. D sets distances. Options: Polyline (P), Angle (A).
ARRAY	AR	Create rectangular, polar, or path arrays of objects. Classic: ARRAYCLASSIC.
STRETCH	S	Stretch geometry by moving endpoints within a crossing selection window.
LENGTHEN	LEN	Lengthen or shorten lines/arcs. Options: Delta (DE), Percent (P), Total (T).
BREAK	BR	Break objects at two points or at a point (BR P). Removes segment between points.
JOIN	J	Join collinear lines, overlapping arcs, and polylines into a single object.
EXPLODE	X	Explode blocks, polylines, dimensions, hatches into constituent elements.
PEDIT	PE	Edit polylines: Join, Width, Fit, Spline, Reverse. Also converts lines to polylines.
SPLINEDIT	SPE	Edit spline control vertices, fit points, open/close, add/remove vertices.
GRIP EDIT	—	Click object to show grips. Click grip then move/stretch/rotate. SPACE cycles modes.

★ Trim/Extend Master Tip

Press ENTER at "Select cutting edges" prompt → ALL objects become cutting edges automatically.
 This is the fastest workflow: TR > ENTER > click everything to trim.
 SHIFT+click during TRIM = extend; SHIFT+click during EXTEND = trim.

SECTION 3
 LAYER · PROPERTIES · FILTERS · STANDARDS

Layer Management

Layers are the organizational backbone of every AutoCAD drawing. Professional drawings use industry-standard layer naming conventions to ensure compatibility, readability, and collaboration across teams and disciplines.

Command	Alias	Description
LAYER	LA	Open Layer Properties Manager. Create, delete, rename, freeze, lock layers.
LAYERSTATE	—	Save and restore complete layer configurations (visibility, color, linetype).
LAYON	—	Turn on all layers. Equivalent to setting all layers to ON state.
LAYOFF	—	Turn off layers by clicking objects on them. Objects invisible but still exist.
LAYFRZ	—	Freeze layers by clicking objects. Frozen layers excluded from REGEN.
LAYTHW	—	Thaw all frozen layers. Reverse of LAYFRZ.
LAYLCK	—	Lock layers by clicking objects. Objects visible but not selectable.
LAYULK	—	Unlock layers by clicking objects.
LAYMCH	—	Match selected objects' layer to destination layer by clicking target.
LAYCUR	—	Move selected objects to current layer.
LAYDEL	—	Delete all objects on a layer and remove the layer from the drawing.
LAYISO	—	Isolate a layer: all other layers faded or locked. LAYUNISO to restore.
LAYWALK	—	Walk through layers: highlights each layer in sequence to check content.
-LAYER	—	Command-line layer management (scriptable, without dialog).

AIA/NCS Layer Naming Standard

The AIA CAD Layer Guidelines (National CAD Standard) defines the format: **L-DDDD-SSSS-MMMM** where L=discipline, D=major category, S=minor, M=status.

Discipline Code	Major Group	Example Layer	Description
A (Architectural)	WALL	A-WALL-FULL	Full-height walls
S (Structural)	COLS	S-COLS-CONC	Concrete columns
M (Mechanical)	DUCT	M-DUCT-SUPP	Supply ductwork
E (Electrical)	LITE	E-LITE-CLIG	Ceiling light fixtures
P (Plumbing)	PIPE	P-PIPE-DOMW	Domestic water piping
C (Civil)	TOPO	C-TOPO-MAJR	Major topographic contours
L (Landscape)	PLNT	L-PLNT-TREE	Trees

G (General)	ANNO	G-ANNO-TEXT	General annotation text
0	—	0 (zero)	Default layer — geometry placed here only temporarily
DEFPOINTS	—	DEFPOINTS	Dimension definition points — never plot

SECTION 4
 LINEAR · ALIGNED · ANGULAR · RADIAL · LEADERS

Dimensioning Commands

Command	Alias	Description
DIMLINEAR	DLI	Horizontal or vertical dimensions. Smart: picks correct orientation.
DIMALIGNED	DAL	Dimensions aligned to an angled object (true length).
DIMANGULAR	DAN	Dimensions the angle between two lines or three points.
DIMRADIUS	DRA	Radius dimension with leader pointing to arc or circle center.
DIMDIAMETER	DDI	Diameter dimension (Ø symbol). Click on circle or arc.
DIMARC	DAR	Measures arc length. Displays arc symbol (⌢) above dimension text.
DIMBASELINE	DBA	Creates dimensions from a common baseline. Continue after first dim.
DIMCONTINUE	DCO	Creates chain of dimensions continuing from previous endpoint.
DIMORDINATE	DOR	Creates ordinate (datum) dimensions showing X or Y coordinates.
QDIM	QDIM	Quick dimension: select objects and click → multiple dims at once.
DIM	DIM	Smart dimensioning: automatically selects correct dim type from selection.
DIMSPACE	—	Equally space parallel linear dimensions at a specified distance.
DIMBREAK	—	Adds breaks to dimension lines where they cross geometry.
DIMEDIT	DED	Edit dimension text or extension line angles. Options: Home, New, Rotate.
DIMSTYLE	D	Manage dimension styles (text height, arrowheads, tolerances, scale).
DIMTEDIT	—	Move or rotate dimension text. Modes: Left, Right, Center, Home, Angle.
QLEADER	LE	Quick leader with text or block annotation.
MLEADER	MLD	Multileader: supports multiple leaders with formatted content.

■ Dimension Style Best Practices

- Always create a project-specific DIMSTYLE — never use the default "Standard" style in production.
- Set DIMSCALE to match your plot scale factor (e.g., 1:100 → DIMSCALE = 100 in Model Space).
- Use ANNOTATIVE dimension style for layouts — AutoCAD handles scale automatically.
- Create separate dimstyles for different annotation scales (1:50, 1:100, 1:200).

SECTION 5

MTEXT · DTEXT · STYLE · FIELD

Annotation & Text

Command	Alias	Description
MTEXT	MT/T	Multiline text with full formatting (bold, italic, color, tabs, lists).
TEXT / DTEXT	DT	Single-line dynamic text. Repositions after each Enter.
SPELL	SP	Check spelling on selected text objects.
FIND	FI	Find and optionally replace text strings within the drawing.
STYLE	ST	Manage text styles (font, height, oblique angle, backwards/upside down).
TEXTTOFRONT	—	Bring all text and dimensions in front of other objects.
SCALETEXT	—	Scale text objects without changing their insertion points.
JUSTIFYTEXT	—	Change text justification without moving the text visually.
ARCTEXT	—	Write text along an arc. (Express Tools required)
FIELD	—	Insert data-linked field text: file name, date, sheet number, object properties.
MLEADER	MLD	Multileader annotation. Supports text, blocks, and complex callouts.
TOLERANCE	TOL	Insert GD&T; (geometric dimensioning and tolerancing) symbols.

SECTION 6

BLOCK · INSERT · XREF · WBLOCK

Blocks, Xrefs & References

Command	Alias	Description
BLOCK	B	Define a block from selected objects. Assigns name, base point, and properties.
INSERT	I	Insert a block definition into the current drawing.
WBLOCK	W	Write a block or objects to a separate DWG file on disk.
EXPLODE	X	Explode a block insert to its constituent elements.
BEDIT	BE	Edit a block definition in the Block Editor environment.
REFEDIT	—	Edit an Xref or nested block in-place within the host drawing.
ATTDEF	ATT	Define an attribute (variable text) to embed in a block.
ATTEDIT	ATE	Edit attribute values in selected block inserts.
EATTEDIT	—	Enhanced attribute editor with spreadsheet-style editing.
ATTEXT	—	Extract attribute data from blocks to CSV/TXT/DXF.
XREF	XR	Manage external reference files: attach, bind, overlay, detach.
XATTACH	XA	Attach an external DWG as an Xref.
XBIND	XB	Permanently bind Xref-dependent symbols (layers, styles) into drawing.
XCLIP	XC	Clip an Xref or block with a boundary (rectangle, polygon, or polyline).
REGEN	RE	Regenerate the drawing to update display of objects and Xrefs.

SECTION 7

ZOOM · PAN · VIEW · VIEWPORTS

View & Navigation

Command	Alias	Description
ZOOM	Z	Zoom options: All (A), Extents (E), Window (W), Previous (P), Scale (S), Realtime.
PAN	P	Pan the drawing view. Press Esc or Enter to exit. Middle mouse button drags.
VIEW	V	Save and restore named views. Useful for returning to key areas quickly.
MVIEW	MV	Create and manage model space viewports in paper space layouts.
VPORTS	—	Create tiled viewports in model space for simultaneous multi-view editing.
VPCLIP	—	Clip a viewport with a custom boundary shape.
VPLAYER	—	Freeze/thaw layers independently in each viewport.
3DORBIT	3DO	Orbit around 3D model in real time. SHIFT+Middle button orbits.
PLAN	—	Return to plan (top) view in 3D. Options: Current UCS, World, Named UCS.
VSCURRENT	VS	Change visual style: 2D Wireframe, Conceptual, Realistic, X-Ray.
REGEN	RE	Regenerate model (recalculate display). Fixes display artifacts.
REGENALL	REA	Regenerate all viewports simultaneously.

SECTION 8

SELECT · FILTER · PROPERTIES · QUICK SELECT

Selection & Properties

Command	Alias	Description
SELECT	SEL	Manually build selection sets. Options: Last, All, Window, Crossing, Fence, WPolygon, CPolygon.
QSELECT	—	Quick Select: filter objects by type, layer, color, or property.
FILTER	FI	Object Selection Filter: complex logical conditions (AND/OR) across all properties.
PROPERTIES	PR / CTRL+1	Display/edit all properties of selected objects.
MATCHPROP	MA	Copy properties (layer, color, linetype, style) from source to target objects.
CHPROP	—	Change properties via command line (Layer, LType, Color, LTScale, Thickness).
SELECTSIMILAR	—	Add all similar objects (same block name, layer, or type) to selection.
GROUP	G	Create a named group of objects for easy re-selection.
UNGROUP	—	Dissolve a group back to individual objects.
ISOLATEOBJECTS	—	Isolate selected objects; hide all others. HISOLATEOBJECTS hides them.
UNISOLATEOBJECTS	—	Restore all hidden/isolated objects to full visibility.

SECTION 9

PLOT · PUBLISH · PDF · CTB/STB

Plotting & Publishing

Command	Alias	Description
PLOT	CTRL+P	Open Plot dialog: printer, paper size, plot area, scale, style table.
PUBLISH	—	Batch-plot multiple layouts or sheets to DWF, DWFx, PDF, or printer.
PAGESETUP	—	Define page setup (printer/plotter, paper size, scale) without plotting.
EXPORTPDF	—	Export drawing to PDF. Supports single sheet or all layouts.
EXPORTDWF	—	Export to Design Web Format for online sharing and review.
ETRANSMIT	—	Package drawing with all dependencies (Xrefs, images, fonts) for transmittal.
RECOVER	—	Recover a damaged DWG file that will not open normally.
AUDIT	—	Audit drawing for errors and fix (AUDIT then answer "Yes" to fix errors).
PURGE	PU	Remove unused blocks, layers, styles, linetypes to reduce file size.
DWGPROPS	—	Set drawing properties (title, subject, author, keywords) for file management.
STATUSBAR	—	Toggle the status bar display (OSNAP, GRID, ORTHO, POLAR, etc.)
PLOTTERMANAGER	—	Open the Plotter Manager to configure plotter/printer PC3 files.
STYLESMANAGER	—	Open Plot Style Manager to edit CTB (Color-dependent) or STB tables.

■ Plot Scale Best Practice

Always plot from PAPER SPACE layouts — not from Model Space directly.
 Set viewport scale (VP scale) = 1:drawing scale (e.g., 1:100). Then plot at 1:1.
 Use ANNOTATIVE text and dims — AutoCAD adjusts scale per viewport automatically.
 Use a project CTB file with consistent lineweights: 0.18mm (thin), 0.35mm (medium), 0.70mm (thick).

SECTION 10

EXTRUDE · REVOLVE · UNION · SUBTRACT

3D Modeling Commands

Command	Alias	Description
EXTRUDE	EXT	Extrude 2D profile to create 3D solid or surface. Supports taper angle.
REVOLVE	REV	Revolve 2D profile around an axis to create solid of revolution.
LOFT	LOFT	Loft between cross-section profiles to create organic 3D shapes.
SWEEP	SWEEP	Sweep a 2D profile along a path curve to create 3D solid or surface.
UNION	UNI	Boolean union: combine two or more 3D solids into one.
SUBTRACT	SU	Boolean subtract: remove one solid from another (holes, pockets).
INTERSECT	IN	Boolean intersect: keep only the overlapping volume of two solids.
CHAMFEREDGE	—	Chamfer edges of 3D solids. Select edges, set distances.
FILLETEGE	—	Fillet (round) edges of 3D solids. Select edges, set radius.
SECTION	SEC	Create 2D cross-section from a 3D solid using a cutting plane.
SLICE	SL	Slice a 3D solid with a plane. Keep one half, both halves, or selected side.
SHELL	—	Create hollow shell from a 3D solid. Specify wall thickness and open faces.
3DARRAY	—	Create 3D rectangular or polar arrays of 3D objects.
3DMIRROR	—	Mirror 3D objects about a plane (XY, YZ, ZX, or 3-point defined plane).
3DROTATE	3R	Rotate 3D objects around X, Y, or Z axis using 3D rotate gizmo.
3DMOVE	3M	Move objects in 3D using move gizmo with axis constraints.
RENDERPRESETS	—	Configure rendering quality and output for 3D visualizations.

SECTION 11

OVERKILL · FLATTEN · BURST · SUPERHATCH

Advanced & Express Tools

Command	Source	Description
OVERKILL	Standard	Remove duplicate and overlapping geometry. Options: Tolerance, Ignore properties.
FLATTEN	Express	Flatten 3D geometry to Z=0 elevation for 2D plotting.
BURST	Express	Explode block and retain attribute values as Mtext objects (not EXPLODE).
SUPERHATCH	Express	Hatch with an image, block, or Xref as the hatch pattern.
TXTEXP	Express	Explode text to polylines (for CNC export, unusual fonts).
ARCTEXT	Express	Write text along an arc path.
BREAKLINE	Express	Insert break line symbol automatically sized to viewport.
DIMREASSOC	Standard	Reassociate dimensions to geometry if they have become disassociated.
FIXDISASSOCIATEDDIM	—	Fix disassociated dimensions using automatic object recognition.
NCOPY	Express	Copy objects nested in Xrefs or blocks to the current space.
MSTRETCH	Express	Stretch multiple objects simultaneously.
TCOUNT	Express	Add sequential numbering to multiple text objects automatically.
CONVERT	Standard	Convert proxy objects (from other software) to native AutoCAD objects.
DRAWORDER	DR	Control draw order: Front, Back, Above/Below Object.
TEXTMASK	Express	Apply a wipe-out mask behind text for clear readability over hatching.
LISP	—	Run AutoLISP routines. Commands: APPLOAD, VLIDE, (load "file.lsp")
ACTRECORD	—	Action Recorder: record commands as a macro for automation.

SECTION 12

Function Keys · Ctrl · System Variables

Keyboard Shortcuts Master List

Keyboard Shortcuts

Shortcut	Action	Shortcut	Action
F1	Help	F2	Text Window (Command History)
F3	OSNAP Toggle	F4	3D Object Snap (Model Space)
F5	Isoplane Toggle	F6	Dynamic UCS Toggle
F7	Grid Display	F8	ORTHO Mode
F9	SNAP Toggle	F10	Polar Tracking
F11	Object Snap Tracking	F12	Dynamic Input
CTRL+A	Select All	CTRL+B	Grid Snap On/Off
CTRL+C	Copy to Clipboard	CTRL+V	Paste from Clipboard
CTRL+SHIFT+V	Paste as Block	CTRL+D	Dynamic UCS
CTRL+E	Isoplane Cycle	CTRL+F	Running OSNAP
CTRL+G	Grid	CTRL+H	PICKSTYLE toggle
CTRL+J / Enter	Repeat Last Command	CTRL+K	Hyperlinks
CTRL+L	ORTHO Mode	CTRL+N	New Drawing
CTRL+O	Open Drawing	CTRL+P	Plot (Print)
CTRL+Q	Exit AutoCAD	CTRL+R	Cycle Viewports
CTRL+S	Save	CTRL+SHIFT+S	Save As
CTRL+T	Tablet Mode	CTRL+U	Polar Tracking
CTRL+W	Object Snap Tracking	CTRL+X	Cut to Clipboard
CTRL+Y	Redo	CTRL+Z	Undo
CTRL+1	Properties Palette	CTRL+2	DesignCenter
CTRL+3	Tool Palettes	CTRL+4	Sheet Set Manager
CTRL+6	dbConnect	CTRL+7	Markup Set Manager
CTRL+8	Quick Calc	CTRL+9	Command Line
ESC	Cancel Command	ENTER/SPACE	Confirm / Repeat Last Cmd

Essential System Variables

Variable	Default	Description
LTSCALE	1.0	Global linetype scale factor. Increase for smaller-scale drawings.
PSLTSCALE	1	0=linetype scale controls both spaces; 1=scale relative to viewport.
DIMSCALE	1.0	Dimension scale factor. Set = 1/plot scale for Model Space dims.
TEXTSIZE	0.2	Default height for single-line text created with the TEXT command.
PICKBOX	3	Size (pixels) of the selection pick box cursor.
PICKFIRST	1	1=select before command (noun-verb); 0=command then select.
PDMODE	0	Point display mode (0=dot, 3=X, 33=X with circle, 35=star with circle).
FILEDIA	1	1=show file dialog boxes; 0=command-line file entry only.
MEASUREMENT	0	0=imperial (inch) hatch/linetype definitions; 1=metric.
SAVETIME	10	Auto-save interval in minutes. Set to 5-10 for safety.

SECTION 13
 NCS · Color Standards · Linetype

Layer Standards & Best Practices

Professional layer standards ensure drawing quality, file compatibility across design teams, and efficient plotting. AIA Layer Guidelines / NCS is the dominant standard in the USA; ISO 13567 is the international equivalent.

Recommended Layer Color Coding

Color	Number	Object Type	Standard Use
White / 7	7	Primary drawing elements	Main geometry (walls, structural elements)
Red / 1	1	Electrical, fire safety	Electrical conduit, fire alarm systems
Yellow / 2	2	Furniture, equipment	FF&E; items, process equipment
Green / 3	3	Plumbing, mechanical	Pipe systems, HVAC ductwork
Cyan / 4	4	Temporary / construction	Construction phasing, demolition
Blue / 5	5	Dimensions, annotation	All dimensioning and notes
Magenta / 6	6	Civil, structural	Survey, structural steel, concrete
Gray / 8	8	Hatching, poche	Background fills, material indication
Dashed / 9	9	Hidden, future work	Hidden lines, future phase elements

■ Layer Best Practices

- NEVER draw on layer 0 (zero) in production — use it only as a temporary scratch layer.
- All objects should be ByLayer for color, linetype, and lineweight — never override per object.
- Create a Layer Template DWT file with all standard layers pre-defined for every new project.
- Use Layer States to save/restore layer configurations for different view exports (floor plan, ceiling plan, etc.).

SECTION 14
 LIMITS · UNITS · TEMPLATE · PAGE SETUP

Drawing Setup & Templates

Command	Description
NEW (with template)	Start new drawing from DWT template with pre-set layers, styles, and settings.
UNITS	Set drawing units (decimal, architectural, engineering), precision, and angle settings.
LIMITS	Set drawing bounds. GRID will display within limits. Not strictly enforced.
MVSETUP	Set up viewport, units, and scale in a single workflow.
PAGESETUP	Configure plotter, paper size, and plot style for each layout tab.
LAYOUT	Create, copy, rename, delete paper space layouts.
LAYOUTWIZARD	Step-by-step layout creation: plotter, paper, viewport, title block.
DWGPROPS	Set drawing metadata: Title, Subject, Author, Keywords for file management.
OPTIONS	Customize all AutoCAD settings: display, file paths, selection, plotting, etc.
CUSTOMIZE	Customize ribbons, toolbars, keyboard shortcuts, and workspaces.
CUI	Complete User Interface customization (ribbons, menus, workspaces, macros).

SECTION 15

Performance · Recovery · Productivity

Troubleshooting & Pro Tips

★ Speed Up AutoCAD

Type CLEANSCREENON / F9 for distraction-free fullscreen drafting.
PURGE regularly to remove unused blocks/styles and reduce file size.
AUDIT after PURGE to fix errors. Restart after WSETTINGS changes.
Turn off hardware acceleration if experiencing display glitches.
Limit the number of Xrefs open simultaneously — each adds overhead.

■ Common Mistakes & Fixes

Nothing on screen: type ZOOM > E (Extents) to find all objects.
Objects not snapping: check F3 (OSNAP) is ON and correct modes enabled.
Dimension text too small/large: check DIMSCALE matches your plot scale.
Layer frozen vs off: frozen layers excluded from REGEN — use freeze for speed.
Hatching fails: objects must form a completely closed boundary (use BOUNDARY).

■ Productivity Hacks

Right-click on ribbon > Customize → add your most-used commands to Quick Access Toolbar.
CTRL+SHIFT+click selects sub-objects within blocks without exploding.
Double-click to enter block editor; double-click text to edit in-place.
BCOUNT (Express) counts block instances — useful for BOM/quantity take-offs.
Use ACTION RECORDER (ACTRECORD) to automate repetitive multi-step tasks.

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